

The MAN-MACHINE MIRACLE



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The man-machine combination concept is an old one. Way back in 1908, Jean de la Hire introduced us to Nyctalope, which was probably the first real superhero and cyborg in his book titled *L'Homme Qui Peut Vivre Dans L'eau* (*The Man Who Can Live in the Water*). In 1928, author Edmond Hamilton told us about space explorers who were partly-organic and partly-machine in his book *The Comet Doom*.

The term cyborg has come from the phrase cybernetic organism, and in layman terms it is a theoretical or fictional being possessing both organic and bio-mechatronic parts. It was created by Manfred Clynes and Nathan S. Kline in 1960 to refer to an enhanced human being who could survive in extraterrestrial environments.

The book *Cyborg: Evolution of the Super-man* by D.S. Halacy released in 1965 talked of a new frontier that was “not merely space, but more profoundly the relationship between inner space to outer space—a bridge between mind and matter,” as mentioned in its introduction.

Cyborgs have also figured in Hollywood movies to a large extent. For example, cybermen in Doctor Who franchise, Borg in Star Trek and Darth Vader in Star Wars. While these looked more mechanical, the terminators from Terminator movie series and the human cyborgs from Battle Star Galactica, among others, appeared more human. Cyborgs in fiction are often portrayed with

Haley Joel Osment played a robot created with the ability to have emotions, dreams and desires in Steven Spielberg's movie *A.I. Artificial Intelligence*, a sci-fi adaptation of Pinocchio story. It takes us into amazingly-rendered future worlds of humans and the robots they create to serve them.

The movie begins in a classroom, with a leisurely discussion on the nature of love. From this lecture is born David (Haley Joel Osment), the first robot created with the ability to have complex emotions, dreams and desires.

Robots do our work, babysit our kids, cure us of diseases and provide sexual pleasure. But David is different. His job is to be someone's little boy. He is necessary because pregnancies are strictly regulated. His role is, essentially, to replace humans!

physical or mental abilities that are much more superior than humans'.

The term cyborg is a little different from the word bionic and often stands for an organism that has restored function or enhanced abilities due to the integration of some artificial component or technology that relies on some sort of feedback. While cyborgs are popularly conceived as mammals, these could even be any sort of organism. The term cybernetic organism has been applied to networks such as road systems, corporations and governments, and has also been extended to micro-organisms that are adapted to perform at higher levels than their unchanged equivalents.

Types of cyborgs

As per a school of thought, humans with any kind of physical attachments, even those using very basic devices such as an artificial cardiac pacemaker or implantable cardioverter-defibrillator to enhance their biological capabilities, can be called cyborgs. This is because these devices measure voltage potentials in the body, perform signal processing and can deliver electrical stimuli, which enables the persons to live.

Fig. 1: Darth Vader in the movie franchise Star Wars looked very mechanical (Image courtesy: www.geeksofdoom.com)

